

University of Southern Denmark

Quantifying Pressing in Professional Football using Tracking Data

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Declaration of Title and Authorship

I declare that this thesis and the work presented in it is my own and has been produced by me as the result of my own original research.

- This work was performed while I was enrolled at the University of Southern Denmark.
- All sources are properly acknowledged.
- All quotations are clearly cited.

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Abstract

Pressing has become an increasingly important tactical concept in modern professional football, yet the quantitative analysis of collective defensive behaviour remains methodologically challenging. Many existing approaches evaluate defensive pressure using isolated events or frame-level spatial measurements, which may struggle to represent the temporal continuity and contextual development of coordinated pressing behaviour.

This thesis presents a contextual sequence-based framework for the detection and evaluation of settled high-pressing sequences using synchronized event and tracking data from professional football. The proposed framework combines spatial pressure modelling, temporal continuity logic, contextual possession analysis, and outcome-based evaluation in order to model pressing as a continuous tactical process rather than as isolated defensive actions.

Tracking data was used to model player positioning, movement, and defensive pressure behaviour, while synchronized event data supplied contextual information regarding possession structure, passing actions, and tactical outcomes. Detected pressing sequences were subsequently evaluated using both outcome classification and a composite press-scoring framework incorporating tactical and behavioural characteristics.

The resulting analyses demonstrated that the framework was capable of identifying tactically plausible pressing sequences exhibiting meaningful spatial and temporal variation throughout the analysed match. Qualitative sequence evaluation further showed that the framework frequently captured coherent sequence initiation, continuation, and termination behaviour. At the same time, the results also highlighted important methodological challenges relating to sequence-boundary ambiguity, fragmentation, parameter sensitivity, and dependence on event-data richness.

Overall, the thesis demonstrates that contextual sequence-based modelling provides a useful and interpretable foundation for football pressing analysis while also illustrating the inherent complexity of discretizing continuous collective defensive behaviour into temporally bounded analytical sequences.

Chapter 1

Introduction

1.1 Background and Motivation

The increasing availability of high-resolution event and tracking data has significantly expanded the role of data-driven analysis within professional football (Gudmundsson & Horton, 2017; Rein & Memmert, 2016). Modern football analytics increasingly seeks to model complex tactical behaviours involving coordinated player movement, spatial control, and collective team dynamics.

Among these tactical behaviours, pressing has become one of the most important defensive concepts in contemporary football (Freitas et al., 2023; Low et al., 2021). High pressing is frequently used to disrupt opposition build-up play, constrain progression options, and create advantageous transition opportunities in advanced areas of the pitch. However, despite its tactical importance, pressing remains difficult to quantify consistently due to its collective and context-dependent nature (Forcher et al., 2024; Herold et al., 2022).

This thesis investigates the detection and evaluation of settled high-pressing behaviour using synchronized event and tracking data.

1.2 Problem Statement

Despite the growing availability of football event and tracking data, the quantitative analysis of collective pressing behaviour remains methodologically challenging (Forcher et al., 2024; Herold et al., 2022). Many existing approaches evaluate defensive pressure using isolated events or frame-level spatial measurements, which may struggle to represent the temporal continuity and contextual development of coordinated pressing sequences (Bekkers, 2024; Merckx et al., 2021).

In practice, pressing behaviour evolves dynamically across multiple player interactions, passes, and possession phases. As a result, defining clear sequence boundaries and evaluating pressing effectiveness consistently remains difficult. Temporary reductions in active pressure, transitional circulation phases, and overlapping defensive actions may all introduce ambiguity regarding whether a pressing sequence should continue or terminate.

Although synchronized event and tracking data provide opportunities for richer contextual analysis, integrating these data sources into coherent sequence-based representations of pressing behaviour remains a complex problem (Gudmundsson & Horton, 2017; Van Roy et al., 2023).

This thesis therefore addresses the problem of detecting and evaluating settled high-pressing behaviour through a contextual sequence-based framework integrating synchronized event and tracking data.

1.3 Research Objectives

The primary objective of this thesis is to develop a contextual framework for the detection and evaluation of settled high-pressing behaviour in professional football using synchronized event and tracking data. More specifically, the thesis aims to:

- Detect settled high-pressing sequences using spatial, temporal, and contextual information derived from tracking and event data
- Model pressing behaviour as a continuous sequence-based tactical process rather than as isolated defensive actions
- Evaluate pressing effectiveness through contextual outcome analysis and composite press scoring
- Produce interpretable visual and analytical representations of collective defensive behaviour in professional football

1.4 Scope and Limitations

This thesis focuses specifically on the detection and evaluation of settled high-pressing behaviour in professional football using synchronized event and tracking data. The proposed framework is designed to analyse sustained defensive pressure occurring during organized opposition build-up phases rather than all forms of defensive activity throughout a match.

The study is exploratory in nature and emphasizes contextual interpretability over large-scale predictive modelling or tactical classification. The framework was evaluated using a single professional football match in order to enable detailed qualitative and sequence-based analysis of pressing behaviour.

Furthermore, the proposed approach does not attempt to provide a universally definitive representation of football pressing, as defensive pressure remains inherently context-dependent and subject to tactical interpretation. Instead, the thesis aims to provide an interpretable and extensible framework for contextual pressing analysis using integrated spatiotemporal football data.

1.5 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 reviews existing literature relating to football pressing analysis, tracking-data analytics, and contextual spatiotemporal modelling in professional football.

Chapter 3 describes the datasets used throughout the study together with the preprocessing and contextual preparation procedures applied prior to analysis.

Chapter 4 presents the proposed methodological framework for detecting and evaluating settled high-pressing sequences using synchronized event and tracking data.

Chapter 5 presents the resulting analyses, visualizations, and qualitative sequence evaluations produced by the framework.

Chapter 6 discusses the broader implications, strengths, limitations, and future development possibilities associated with the proposed approach.

Finally, Chapter 7 concludes the thesis by summarizing the main findings and contributions of the study.

Chapter 2

Literature Review

2.1 Pressing as a Tactical Concept

Pressing has become one of the defining tactical characteristics of modern professional football (Freitas et al., 2023; Low et al., 2021). Broadly, pressing refers to coordinated defensive actions intended to restrict an opponent's time, space, and ability to progress possession. Rather than defending passively in fixed positions, pressing teams actively engage opponents by closing passing lanes, applying pressure to the ball carrier, and forcing hurried decisions or turnovers. As football has evolved toward faster ball circulation and structured build-up play, pressing has increasingly been used not only as a defensive mechanism but also as a means of creating attacking opportunities through regaining possession in advanced areas of the pitch.

Although pressing is often discussed as a single concept, it encompasses multiple tactical forms that differ substantially in structure and intent (Low et al., 2018, 2021). High pressing involves applying defensive pressure close to the opponent's goal during their build-up phase, with the objective of disrupting possession before the attacking team can progress up the field. Mid- and low-block defensive systems, by contrast, apply pressure in deeper areas while prioritizing compactness and space protection. Another related concept is counter-pressing, often referred to as *Gegenpressing*, in which immediate pressure is applied following loss of possession in an attempt to regain the ball before the opponent can organize their attack.

The present thesis focuses specifically on **settled high pressing**, rather than counter-pressing or deeper defensive pressure. Settled high pressing refers to organized defensive pressure applied against an opponent in controlled possession within their defensive third or early build-up phase. In these situations, the defending team is already positioned high up the pitch and attempts to constrain the opponent's progression through coordinated movement and pressure on likely passing options. This form of pressing differs conceptually from transitional defensive actions because it occurs within relatively stable possession structures rather than immediately after turnovers.

From an analytical perspective, pressing presents several challenges. Unlike discrete football events such as passes or shots, pressing is inherently continuous and collective in nature. It emerges through coordinated movement, spatial positioning, and evolving interactions between multiple players over time. As a result, pressing is difficult to represent using traditional

event-based datasets alone. Simple metrics such as tackles or interceptions capture only isolated defensive outcomes and fail to describe the broader tactical process through which pressure is applied.

Modern tracking data provide new opportunities for analysing pressing behaviour in greater detail. By recording the continuous movement of players and the ball, tracking systems enable researchers to study defensive pressure as a dynamic spatiotemporal process rather than a collection of isolated events. This allows pressing to be examined in relation to movement intensity, directional behaviour, spatial pressure, and temporal continuity.

Despite growing interest in pressing analysis, much existing work remains focused either on aggregate team statistics or on simplified frame-level pressure indicators. Comparatively less attention has been devoted to modelling pressing as a context-dependent sequence of coordinated defensive behaviour. This limitation is particularly relevant for settled high pressing, where tactical meaning depends heavily on possession stability, spatial context, and sustained defensive engagement.

Accordingly, the present thesis approaches pressing as a dynamic and context-sensitive tactical process. The following sections review the analytical methods used to quantify football behaviour using spatiotemporal data and position the proposed framework within the broader literature on sequence-based football analysis.

2.2 Quantifying Football Performance with Spatiotemporal Data

The increasing availability of football data has significantly expanded the scope of quantitative performance analysis (Gudmundsson & Horton, 2017; Rein & Memmert, 2016). Early analytical approaches relied primarily on event-based datasets, which record discrete match actions such as passes, shots, tackles, and interceptions. These data enabled the development of influential metrics including expected goals (xG), passing networks, possession statistics, and defensive indicators such as Passes Per Defensive Action (PPDA). Such approaches have contributed substantially to the understanding of team performance and tactical behaviour by transforming football events into measurable analytical outputs.

Despite their usefulness, event-based datasets have important limitations when analysing pressing behaviour (Forcher et al., 2024; Herold et al., 2022). Events capture only isolated moments in a match and provide limited information about off-ball movement, defensive positioning, or the continuous interaction between players. As a result, many aspects of defensive pressure remain difficult to quantify using event data alone. For example, a defender's ability to constrain passing options or force hurried decisions may not generate an observable event despite having significant tactical impact.

The emergence of optical and GPS-based tracking systems has therefore marked an important development in football analytics (Bialkowski et al., 2014; Gudmundsson & Horton, 2017). Tracking data record the continuous spatiotemporal positions of players and the ball throughout

a match, enabling analysis of movement patterns, spatial relationships, and team structure at frame-level resolution. This allows football behaviour to be studied as a dynamic process rather than a sequence of isolated actions.

Tracking data have supported the development of a wide range of spatial and movement-based analytical models. Existing research has examined concepts such as player influence regions, pitch control, spatial occupation, passing lane obstruction, and team compactness (Fernandez & Bornn, 2018; Spearman et al., 2017). These approaches aim to capture how players interact with space and with one another during different tactical phases of play. In the context of defensive analysis, tracking data enable pressure to be examined not only through observable outcomes, but also through movement intensity, proximity, and spatial constraint.

A central advantage of spatiotemporal data is the ability to analyse off-ball behaviour (Herold et al., 2022). Many tactically important actions in football occur away from the ball and are therefore largely invisible in event-based datasets. Pressing behaviour, in particular, often depends on coordinated movement by multiple defenders attempting to reduce space and restrict progression options. Tracking data make it possible to study these interactions continuously over time and within their broader tactical context.

At the same time, the increased richness of tracking data introduces methodological challenges (Rein & Memmert, 2016; Van Roy et al., 2023). Continuous positional data are substantially more complex than event logs and require preprocessing, synchronization, and contextual interpretation before meaningful tactical inference can be performed. Moreover, the availability of positional information alone does not directly identify concepts such as pressing, possession, or defensive intent. These behaviours must instead be inferred from movement patterns and spatial relationships.

Consequently, recent football analytics research has increasingly shifted toward dynamic and context-aware modelling approaches capable of representing continuous tactical behaviour (Calabuig et al., 2024; Gudmundsson & Horton, 2017). This shift is particularly relevant for pressing analysis, where meaningful interpretation depends not only on player location, but also on movement direction, possession state, temporal continuity, and spatial context. The following section therefore examines sequence-based and dynamic approaches to football analysis that are more closely aligned with the continuous nature of pressing behaviour.

2.3 Dynamic and Sequence-Based Analysis in Football

Football is inherently a dynamic and continuous sport in which tactical behaviour emerges through evolving interactions between players over time (Gudmundsson & Horton, 2017; Rein & Memmert, 2016). Although event-based datasets divide matches into discrete actions, many important tactical processes unfold gradually and cannot be fully understood through isolated events alone. This is particularly relevant for defensive behaviours such as pressing, where coordinated movement, positional adaptation, and sustained pressure develop across sequences of play rather than at single moments.

As football analytics has evolved, increasing attention has therefore been given to temporal and sequence-based approaches capable of representing continuous match dynamics (Bialkowski et al., 2014; Gudmundsson & Horton, 2017). Rather than analysing isolated actions independently, these approaches treat football as a progression of interconnected states in which player behaviour and tactical structure change over time. This shift has supported the analysis of possession phases, attacking build-up sequences, transition moments, and coordinated team movement.

In the context of defensive analysis, sequence-based modelling is especially important because pressing behaviour is rarely instantaneous (Forcher et al., 2024; Herold et al., 2022). A pressing action often begins before direct engagement with the ball carrier and may continue across multiple passes or movement phases. Defenders adjust positioning, close passing lanes, coordinate movement with teammates, and react to anticipated passing options. As a result, pressing cannot be adequately represented by isolated proximity measures or single-frame pressure indicators alone.

Tracking data have enabled researchers to study these dynamics at much finer temporal resolution (Bialkowski et al., 2014; Linke et al., 2020). Continuous positional information makes it possible to analyse movement persistence, acceleration patterns, directional behaviour, and evolving spatial relationships between attackers and defenders. This allows defensive pressure to be interpreted as a process that develops over time rather than a binary event occurring at a single frame.

Several recent studies have consequently adopted dynamic approaches to modelling football behaviour. These include possession-chain analysis, passing sequence evaluation, transition modelling, and frameworks based on temporally connected tactical phases (Aközbek, 2019; Calabuig et al., 2024; Spearman et al., 2017). Such approaches emphasize continuity and interaction rather than isolated match events, reflecting the fluid nature of football itself.

The temporal dimension is particularly important for pressing because defensive intent is often expressed through sustained movement patterns rather than through direct ball actions (Forcher et al., 2024; Low et al., 2021). For example, a defender sprinting toward an anticipated receiver may exert meaningful pressure even before the ball arrives. Similarly, coordinated pressing by multiple players may constrain attacking options without immediately producing a tackle or interception. Analysing pressing therefore requires models capable of representing persistence, anticipation, and sequence continuity.

Despite these developments, many existing pressure metrics remain largely frame-based or outcome-oriented (Bekkers, 2024; Merckx et al., 2021; Robberechts, 2019). Some approaches classify pressure using fixed spatial thresholds around the ball carrier, while others rely primarily on discrete defensive events such as tackles or duels. Although useful in certain contexts, these methods often struggle to capture the evolving and context-dependent nature of pressing behaviour.

Accordingly, the present thesis adopts a sequence-based perspective in which pressing is modelled as a temporally continuous tactical process. Rather than identifying isolated defensive actions, the proposed framework detects sustained pressing runs and aggregates them into

broader pressing sequences occurring within stable tactical contexts. This approach aligns more closely with the continuous and coordinated nature of pressing observed in professional football.

2.4 Contextual Modelling of Pressing

A central challenge in pressing analysis is that defensive pressure cannot be interpreted independently of tactical context (Forcher et al., 2024; Herold et al., 2022). The same defensive movement may represent aggressive pressing in one situation but routine positional adjustment in another. Meaningful quantification of pressing therefore requires consideration not only of player movement, but also of the surrounding game state in which that movement occurs.

One important contextual factor is possession stability. Pressing behaviour is most tactically meaningful when the opposition is in controlled possession, as defensive pressure is typically intended to disrupt structured build-up play or constrain progression options. In unstable situations such as loose balls, aerial duels, or deflections, player movement may become reactive and chaotic rather than strategically coordinated. Consequently, several recent approaches to defensive modelling have emphasized the importance of distinguishing between controlled and uncontrolled phases of play (Calabuig et al., 2024; Herold et al., 2022).

Spatial context is also critical (Fernandez & Bornn, 2018; Spearman et al., 2017). Defensive pressure applied near the opponent's goal differs substantially from pressure applied in deeper defensive areas. High pressing is specifically associated with advanced defensive engagement during the opponent's build-up phase, where teams attempt to force turnovers close to goal. The tactical meaning of pressing therefore depends heavily on pitch location and attacking orientation. Models that ignore spatial context risk treating fundamentally different defensive behaviours as equivalent.

Another important consideration concerns temporal continuity (Forcher et al., 2024; Gudmundsson & Horton, 2017). Pressing sequences often persist through passing exchanges, short transitions, and brief interruptions in movement intensity. However, many simplified pressure metrics evaluate individual frames independently and therefore fail to capture the continuity of coordinated defensive pressure. Context-aware modelling approaches instead recognize that pressing behaviour develops across connected phases of play rather than isolated moments.

The interpretation of pressing is further complicated by the dynamic nature of football possession. During passing phases, for example, the ball may temporarily have no clear carrier while still remaining within a stable possession sequence. Similarly, defensive players may apply pressure toward anticipated receivers before the ball arrives. These situations require models capable of incorporating anticipation and short-term temporal context rather than relying solely on instantaneous positional relationships (Herold et al., 2022; Spearman et al., 2017).

Despite increasing interest in pressure modelling, many existing approaches remain limited by simplified contextual assumptions (Bekkers, 2024; Merckx et al., 2021; Robberechts, 2019). Some methods classify pressure primarily using fixed distance thresholds around the

ball carrier, while others rely on event-derived defensive actions. Although computationally convenient, such approaches may struggle to distinguish tactically meaningful pressing from ordinary defensive movement, particularly during unstable or transitional phases.

Accordingly, recent football analytics research has increasingly emphasized the importance of context-sensitive modelling frameworks that combine spatial, temporal, and possession-based information (Calabuig et al., 2024; Forcher et al., 2024; Herold et al., 2022). Within this perspective, pressing is understood not as a single observable event, but as a tactical process shaped by game state, pitch location, movement continuity, and possession structure.

The present thesis adopts this context-aware perspective by restricting pressing analysis to stable and tactically meaningful game states. In particular, the framework focuses on settled high pressing during controlled opposition possession while excluding unstable aerial phases, loose-ball situations, and transitional counter-pressing contexts. This provides a more precise operational definition of pressing and establishes the foundation for the sequence-based detection framework introduced in the following chapter.

2.5 Research Gap and Thesis Positioning

The growing availability of spatiotemporal tracking data has significantly expanded the possibilities for analysing defensive behaviour in football (Bialkowski et al., 2014; Gudmundsson & Horton, 2017). Existing research has introduced a variety of approaches for measuring pressure, spatial control, player interaction, and defensive intensity. Event-based metrics such as PPDA have provided accessible indicators of pressing activity at team level, while tracking-data studies have enabled more detailed analysis of movement patterns and spatial relationships during defensive phases.

Despite these developments, several limitations remain in the current literature on pressing analysis. First, many existing approaches rely heavily on discrete events or frame-level spatial thresholds (Bekkers, 2024; Merckx et al., 2021; Robberechts, 2019). Although such methods can identify moments of defensive proximity or pressure, they often struggle to represent pressing as a continuous tactical process unfolding over time. Pressing behaviour is inherently dynamic and coordinated, yet many models evaluate defensive pressure independently at each frame without incorporating sequence continuity or sustained behavioural intent.

Second, a substantial proportion of existing work treats pressure in relatively generic terms without clearly distinguishing between different tactical contexts (Forcher et al., 2024; Herold et al., 2022). In practice, pressing behaviour varies considerably depending on possession stability, field position, game phase, and defensive organization. High pressing during structured opposition build-up differs fundamentally from counter-pressing immediately following a turnover or defensive pressure applied in deeper areas of the pitch. Models that do not explicitly account for these contextual differences may therefore combine tactically distinct behaviours under a single definition of pressure.

Third, relatively limited attention has been devoted to modelling pressing at the level of

coordinated pressing sequences (Aközbek, 2019; Calabuig et al., 2024). Existing pressure indicators often focus on isolated defensive actions, individual player proximity, or aggregate team statistics rather than temporally connected phases of sustained pressure. As a result, many current approaches struggle to capture how pressing develops, persists, and terminates within realistic match contexts.

Another limitation concerns the interpretation of pressing outcomes. While some existing metrics quantify the intensity or frequency of pressure, fewer approaches attempt to evaluate the effectiveness or tactical consequences of pressing sequences (Merckx et al., 2021; Robberechts, 2019). This restricts the ability to distinguish between defensive actions that successfully disrupt possession and those that fail to meaningfully constrain attacking progression.

The present thesis addresses these limitations through the development of a context-aware and sequence-based framework for detecting and evaluating settled high pressing using synchronized tracking and event data. Rather than defining pressure through isolated spatial thresholds alone, the proposed approach models pressing as a sustained behavioural process characterized by coordinated movement, directional pursuit, and temporal continuity.

A central contribution of the thesis is the explicit contextual definition of settled high pressing. Pressing detection is restricted to stable opposition possession within advanced build-up areas while excluding aerial phases, loose-ball situations, and transitional counter-pressing contexts. This allows the framework to focus specifically on tactically meaningful high pressing behaviour rather than generic defensive activity.

The framework further contributes through the construction of temporally connected pressing sequences derived from individual pressing runs. By incorporating movement persistence, contextual continuity, and pass-transition handling, the model captures pressing as a dynamic tactical phase rather than a collection of independent moments.

Finally, the thesis introduces an outcome-oriented evaluation perspective in which detected pressing sequences are classified and scored according to their tactical consequences. This extends beyond simple pressure identification by examining whether pressing sequences successfully disrupt or regain possession.

Overall, the thesis positions itself at the intersection of spatiotemporal football analytics, contextual tactical modelling, and sequence-based behavioural analysis. The following chapter presents the methodological framework developed to operationalize these concepts using synchronized tracking and event data from professional football matches.

Chapter 3

Data and Preprocessing

3.1 Data Sources

The analytical framework developed in this thesis is based on professional football tracking and event data provided through a collaboration involving the Danish Football Association (DBU). The dataset contains synchronized match information from multiple professional fixtures and combines high-frequency positional tracking data with structured event annotations and match metadata (Bialkowski et al., 2014; Gudmundsson & Horton, 2017). A subset of matches was additionally provided with separate video footage for qualitative tactical inspection and validation.

The framework processes one match at a time. Although multiple matches are available, match-level processing improves interpretability and allows pressing behaviour to be analysed within its specific tactical and contextual environment. This also reduces computational complexity associated with high-frequency spatiotemporal data processing.

3.1.1 Tracking Data

The primary source of information is the optical tracking dataset, which records the continuous positions of all players and the ball throughout each match (Linke et al., 2020). Positional data are sampled at a frequency of 25 frames per second, producing a detailed spatiotemporal representation of match play.

Each frame contains planar player and ball coordinates together with additional movement-related information such as player speed and ball height. The tracking data therefore capture both on-ball and off-ball movement continuously over time, making them particularly suitable for analysing coordinated tactical behaviour such as pressing.

Unlike event-based datasets, tracking data allow defensive pressure to be examined as a dynamic process involving movement intensity, positional adaptation, and spatial interaction between players. This is especially important for the present study, where pressing behaviour is modelled as a sequence-based and context-dependent defensive phenomenon rather than as isolated defensive actions.

3.1.2 Event Data

In addition to positional tracking data, the dataset includes structured event logs containing timestamped football actions such as passes, shots, fouls, stoppages, and out-of-play events. These events provide important contextual information that complements the continuous positional information contained in the tracking feed.

Although event data alone are insufficient for modelling pressing behaviour directly, they play an important supporting role within the analytical pipeline. In particular, event annotations are used to assist synchronization, identify possession transitions, determine sequence outcomes, and validate important tactical moments observed within the tracking data.

Because the tracking and event feeds originate from separate systems, temporal synchronization between the two sources is required before integrated analysis can be performed. The synchronization procedure used in the present framework is described in the following section.

3.1.3 Match Metadata and Video Footage

Additional contextual information is provided through match metadata associated with each fixture. These metadata include period structure, team identifiers, and attacking orientation for each half. Attacking direction information is particularly important for interpreting spatial relationships consistently across periods and for defining the contextual pressing zone used later in the framework.

A subset of the available matches was additionally accompanied by separate video footage. The video files are not synchronized directly with the tracking feed and are therefore not used for automated analytical processing. Instead, they serve as a qualitative validation resource for manually inspecting detected pressing sequences and verifying the tactical plausibility of model outputs.

Together, the tracking data, event annotations, metadata, and supplementary video footage provide a comprehensive spatiotemporal representation of professional football matches suitable for sequence-based pressing analysis.

3.2 Tracking–Event Synchronization

The tracking and event datasets used in the present study originate from separate recording systems and are therefore not inherently aligned temporally. Because the analytical framework combines continuous positional information with contextual event annotations, synchronization between the two data sources is required before integrated analysis can be performed.

To support synchronization between the two data sources, the framework adopts principles from the **Event and Tracking SYnchronization (ETSY) methodology** (Van Roy et al., 2023), which aligns discrete event annotations with high-frequency tracking data through temporal anchor matching and event-specific alignment procedures. ETSY provides a structured foundation for integrating continuous positional information with contextual event records while

accounting for timing discrepancies between independently recorded data streams.

Even relatively small timing discrepancies can distort frame-level interpretation of possession, movement, and tactical behaviour. A multi-stage synchronization procedure was therefore implemented to align event timestamps with the tracking timeline as accurately as possible.

3.2.1 Global Temporal Alignment

The synchronization process begins with a global alignment step designed to establish a shared temporal reference between the tracking and event feeds. The opening kickoff is used as the primary synchronization anchor because it represents a clearly identifiable event in both datasets.

Within the tracking feed, the kickoff moment is identified through sudden changes in ball movement immediately following the start of play. This moment is then matched to the corresponding kickoff event recorded in the event dataset. The resulting temporal offset is subsequently applied to all event timestamps, producing an initial global alignment between the two data sources.

This alignment step ensures that both datasets operate within a common temporal framework before more detailed frame-level matching is performed.

3.2.2 Event-to-Frame Matching

Following global alignment, individual event records are associated with the most plausible tracking frames. Because event timestamps and tracking frames do not necessarily coincide exactly, local matching procedures are used to identify the best frame-level correspondence for each event type.

For ball-contact events such as passes and shots, local peaks in ball speed are used to approximate the moment of ball interaction within a bounded temporal search window surrounding the aligned event timestamp. This approach improves alignment precision by exploiting the characteristic movement patterns associated with kicking actions.

For event types without distinctive ball-motion signatures, including fouls, stoppages, and out-of-play events, the nearest available tracking frame is selected after temporal offset correction. These events primarily serve contextual rather than movement-based analytical purposes and therefore require less precise spatial alignment.

The resulting matching process produces a frame-level mapping between continuous positional data and discrete event annotations.

3.2.3 Conflict Resolution and Validation

During synchronization, situations may arise in which multiple event records compete for the same tracking frame or where ambiguities exist within the local matching window. To address these cases, a priority-based conflict resolution mechanism is applied.

Higher-impact events such as shots, fouls, or possession-changing actions are given precedence over lower-priority contextual events. This reduces ambiguity in situations involving closely spaced event timestamps and improves consistency within the synchronized dataset.

Following synchronization, a series of validation checks are performed to identify implausible temporal relationships or inconsistent event ordering. Manual inspection of selected sequences is additionally conducted using available video footage and tracking playback in order to verify the overall plausibility of the alignment process.

The final synchronized dataset combines continuous spatiotemporal tracking information with contextual event annotations within a unified frame-level representation of match play. This integrated structure forms the basis for the preprocessing and possession inference procedures described in the following sections.

3.3 Data Cleaning and Standardization

Following synchronization, the tracking and event data were prepared for analysis through a series of cleaning, restructuring, and standardization procedures. These preprocessing steps were necessary to ensure that positional, temporal, and contextual information could be interpreted consistently throughout the analytical pipeline.

Because the framework relies heavily on repeated frame-level spatial and temporal calculations, preprocessing additionally aimed to improve computational efficiency and reduce inconsistencies arising from incomplete or unstable observations.

3.3.1 Coordinate Representation

All positional information was interpreted relative to a fixed pitch coordinate system measured in metres. Player and ball positions were represented using two-dimensional planar coordinates describing horizontal and vertical pitch location, while ball height was retained separately for contextual filtering purposes.

To support consistent spatial interpretation across both halves of a match, all positional calculations were standardized relative to team attacking direction (Gudmundsson & Horton, 2017). This ensured that directional movement, progression toward goal, and spatial pressing zones could be evaluated consistently regardless of period or team orientation.

The resulting coordinate representation provided a unified spatial framework for subsequent movement analysis and tactical interpretation.

3.3.2 Frame-Level Data Structures

The synchronized tracking and event data were reorganized into frame-level analytical structures designed to support repeated spatiotemporal calculations efficiently. Core variables such as frame index, match period, timestamps, player positions, player speeds, and ball coordinates were converted into standardized numeric formats suitable for chronological processing.

Player and ball information were additionally separated into simplified lookup structures to reduce repeated computational overhead during distance calculations, movement analysis,

and possession inference procedures. This restructuring improved the efficiency of repeated player-to-player and player-to-ball operations used throughout the framework.

Event annotations synchronized to individual tracking frames were similarly integrated into the frame-level representation, allowing positional information and contextual events to be analysed within a common temporal structure.

3.3.3 Handling Missing or Invalid States

Certain frames within the tracking data contained incomplete or unstable positional information resulting from temporary tracking loss, ambiguous ball states, or irregular match situations. These frames were handled conservatively in order to reduce the likelihood of introducing erroneous tactical interpretations.

Frames containing missing positional information for the ball or large numbers of players were excluded from analyses requiring reliable spatial reference. Similarly, frames associated with stoppages, out-of-play situations, or unstable tracking states were not considered valid for possession or pressing analysis.

Ball height information was also incorporated into preprocessing in order to distinguish controllable grounded possession from aerial or unstable ball states. Frames involving elevated ball trajectories beyond a predefined threshold were treated separately during later contextual filtering procedures.

The resulting cleaned and standardized dataset provided a stable frame-level spatiotemporal representation of match play suitable for possession inference and sequence-based pressing analysis.

3.4 Inference of Attacking Direction and Possession

Because tracking data do not explicitly encode tactical context such as possession state or attacking orientation, several additional inference procedures were required before pressing behaviour could be analysed reliably (Gudmundsson & Horton, 2017; Linke et al., 2020). These procedures establish a consistent interpretation of directional movement, identify the active ball carrier, determine team possession, and maintain possession continuity during passing phases.

Together, these inferred contextual states form an important foundation for the sequence-based pressing framework developed later in the thesis.

3.4.1 Attacking Direction Inference

Consistent interpretation of movement direction and spatial progression requires knowledge of the attacking orientation of each team during each half of the match. Rather than inferring attacking direction directly from player movement, the framework uses metadata provided alongside the tracking and event datasets.

The metadata specify the attacking orientation of the home team for each period, allowing the attacking direction of both teams to be determined consistently across the match. This information is subsequently used to standardize directional calculations, spatial progression measures, and contextual pressing zones.

Using metadata-based orientation assignment improves robustness and avoids ambiguities that may arise when attempting to infer attacking direction dynamically from positional behaviour alone.

3.4.2 Ball Carrier Identification

Because ball possession is not explicitly encoded within the tracking feed, the framework infers the active ball carrier using spatial proximity between players and the ball.

For each frame, the player nearest to the ball is identified as a potential possession candidate. A player is assigned as the active ball carrier only if the distance to the ball remains below a predefined control threshold consistent with realistic possession distance. This prevents distant or unrelated players from being incorrectly classified as controlling the ball.

To reduce instability caused by brief fluctuations in positional proximity, the framework additionally incorporates a temporal persistence condition. A candidate player must remain the closest valid controller across a minimum number of consecutive frames before possession is formally assigned. This reduces noise arising from momentary tracking variation and improves continuity within inferred possession sequences.

Frames involving unstable or ambiguous ball states are handled conservatively and may remain temporarily unassigned until reliable control conditions are re-established.

3.4.3 Possession Team Assignment

Once an active ball carrier has been identified, possession is assigned to the corresponding player's team. Team-level possession states are subsequently tracked continuously across the match in order to identify possession changes and stable attacking phases.

Possession transitions are detected whenever the inferred ball carrier changes between opposing teams. These transitions play an important role within the broader analytical pipeline, particularly for contextual filtering, pressing sequence termination, and outcome classification procedures introduced later in the thesis.

The framework distinguishes between stable controlled possession and unstable possession states. Frames associated with aerial duels, loose-ball situations, or ambiguous control conditions are not treated as valid possession phases for pressing analysis purposes.

3.4.4 Anticipated Possession During Passing Phases

During grounded passing sequences, the ball may temporarily have no confirmed carrier while travelling between players. However, tactical possession often remains stable throughout these short transition periods despite the absence of direct ball control.

To preserve continuity during passing phases, the framework incorporates an anticipated possession procedure. When no immediate carrier is identified, a limited temporal lookahead window is used to determine the most likely receiving player based on future ball proximity and movement patterns.

The anticipated receiver is then treated as the temporary possession target during the ball's transit phase. This allows defensive pressure and movement behaviour to be evaluated continuously across normal passing exchanges without artificially fragmenting possession or pressing sequences.

By incorporating anticipated possession, the framework better reflects the continuous and dynamic nature of structured football possession during build-up play.

3.5 Contextual State Construction

Following possession inference, additional contextual states were constructed in order to identify situations suitable for pressing analysis. Because the present study focuses specifically on settled high pressing during controlled opposition build-up, not all match situations are considered tactically meaningful for inclusion within the analytical framework.

The preprocessing procedures described in this section establish the contextual filters used to distinguish valid pressing environments from unstable or non-pressable game states.

3.5.1 Ball Height Filtering

Ball height information was incorporated into the preprocessing pipeline in order to distinguish controllable grounded possession from aerial or unstable ball situations. High airborne ball states frequently occur during clearances, long passes, aerial duels, or deflections, where tactical structure and controlled possession may temporarily break down.

To reduce contextual ambiguity, frames in which the ball exceeded a predefined height threshold were excluded from valid pressing analysis states. These situations were treated as non-pressable because defensive movement during aerial phases is often reactive and less representative of coordinated high pressing behaviour.

Filtering aerial states additionally improved the reliability of possession inference procedures by reducing unstable or rapidly changing carrier assignments during airborne ball movement.

3.5.2 Pressing Zone Definition

Because the framework focuses specifically on settled high pressing, defensive pressure is only analysed within advanced opposition build-up areas of the pitch. A contextual pressing zone was therefore defined relative to the attacking team's own goal.

The pressing zone represents the spatial region in which opposition possession is considered vulnerable to organized high pressing behaviour. Frames in which the attacking team pro-

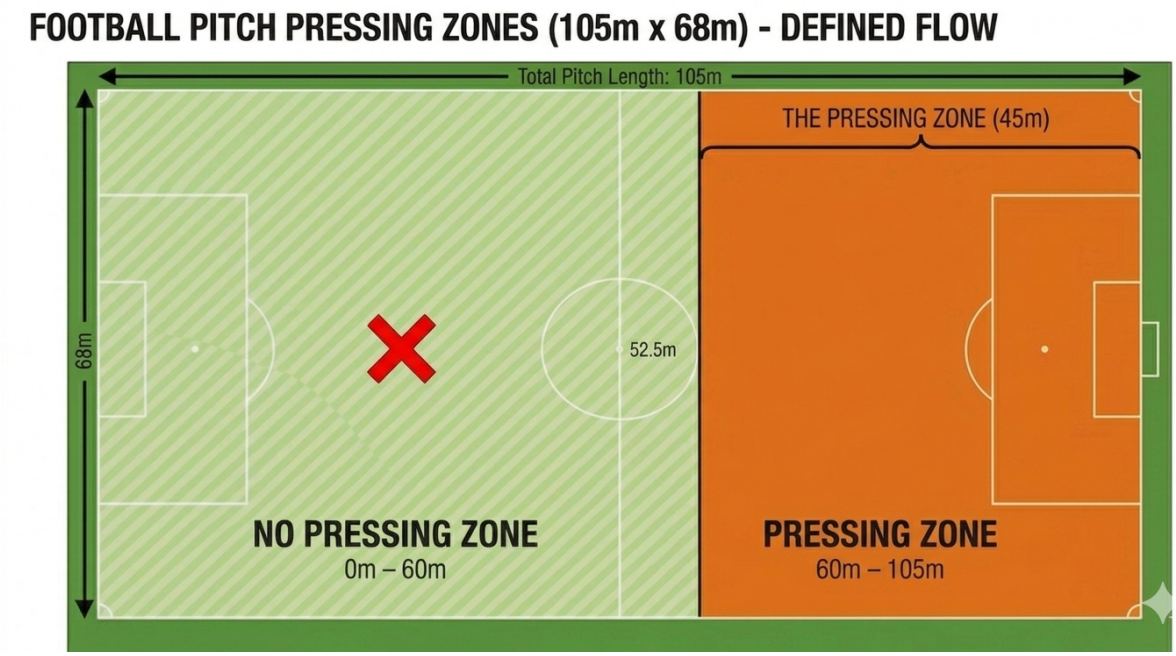


Figure 3.1: Pressing Zone

gressed substantially beyond this zone were excluded from later pressing analysis procedures, as defensive engagement in deeper attacking areas may reflect different tactical structures and objectives.

The pressing zone is evaluated dynamically relative to attacking direction in each half, ensuring that spatial filtering remains consistent across both periods of play. Rather than relying on fixed pitch coordinates, the framework evaluates pressure relative to the attacking team's own goal location. This allows the pressing zone to remain tactically consistent despite directional changes between halves.

For visualization and analytical consistency, pressing contexts are directionally normalized such that the defending team is always oriented from left to right across the pitch. Consequently, the pressing team is interpreted as defending the left side of the pitch while applying pressure toward the opposition goal on the right. Figure 3.X illustrates the conceptual pressing zone used throughout the framework following this normalization procedure.

The contextual pressing-zone restriction plays an important role within the broader analytical pipeline by limiting sequence detection to tactically meaningful high-pressure environments. This filtering process helps distinguish organized settled pressing behaviour from defensive activity occurring in deeper or less structured areas of the pitch.

3.5.3 Identification of Valid Pressing Contexts

Using the contextual states established during preprocessing, the framework identifies frames representing valid settled pressing situations. A frame is considered eligible for pressing analysis only when multiple contextual conditions are simultaneously satisfied.

These conditions include:

- stable team possession,

- controllable grounded ball states,
- valid spatial positioning within the pressing zone,
- and the absence of unstable transitional situations.

Several game states are explicitly excluded from valid pressing contexts. These include loose-ball situations, aerial duels, stoppages, out-of-play periods, and unstable possession transitions immediately following turnovers. Such situations often involve reactive or chaotic movement patterns that differ substantially from the organized defensive behaviour associated with settled high pressing.

The resulting contextual filtering process produces a refined set of tactically meaningful frames suitable for the behavioural pressing analysis introduced in the following chapter.

Chapter 4

Methodology

4.1 Defining Settled High Pressing

The present study focuses specifically on **settled high pressing**, defined as organized defensive pressure applied during controlled opposition possession within advanced build-up areas of the pitch. This definition distinguishes the targeted behaviour from other forms of defensive pressure, such as counter-pressing immediately following possession loss or deep defensive pressure near a team's own goal.

Rather than treating pressing as a generic defensive action, the framework models settled high pressing as a context-dependent tactical process characterized by coordinated forward-oriented defensive engagement during stable opposition possession.

4.1.1 Tactical Definition

Within the proposed framework, pressing is interpreted as an intentional attempt to disrupt or constrain opposition possession through active defensive pressure directed toward the current or anticipated ball carrier. This pressure may involve direct pursuit of the ball carrier, restriction of passing options, or coordinated movement designed to reduce the opponent's available space and decision time.

The framework focuses specifically on high pressing situations occurring during structured opposition build-up phases. These situations are tactically distinct because defending teams attempt to apply pressure high up the pitch in order to disrupt progression close to the opponent's goal and potentially create advantageous turnover opportunities.

Pressing is therefore treated not as an isolated event, but as a sustained behavioural process involving movement continuity, spatial interaction, and coordinated defensive intent.

4.1.2 Included and Excluded Game States

Not all defensive pressure situations are considered valid settled high pressing contexts. The framework therefore applies several contextual restrictions in order to isolate tactically meaningful high pressing behaviour.

Included situations are limited to phases in which:

- the opposition maintains stable controlled possession,
- the ball remains in a controllable grounded state,
- and play occurs within the predefined pressing zone associated with opposition build-up activity.

Several game states are explicitly excluded from the analysis. These include:

- counter-pressing immediately following turnovers,
- deep defensive pressure near a team's own goal,
- aerial duels,
- loose-ball situations,
- stoppages,
- and out-of-play phases.

These exclusions reduce contextual ambiguity and help distinguish organized high pressing from reactive or unstable defensive movement patterns.

4.1.3 Spatial and Temporal Context

The framework additionally incorporates both spatial and temporal context into the operational definition of pressing. Spatially, defensive pressure is evaluated relative to the attacking team's progression within the pressing zone established during preprocessing. This ensures that pressing behaviour is interpreted consistently across both halves of the match.

Temporally, pressing is treated as a continuous tactical phenomenon rather than a collection of independent frame-level actions. Short interruptions in movement intensity or ball control may occur naturally during passing exchanges and coordinated defensive adjustment. The framework therefore allows limited temporal continuity during grounded passing phases and brief fluctuations in defensive behaviour.

The resulting context-aware definition establishes the conceptual foundation for the behavioural detection procedures introduced in the following sections. By restricting analysis to stable and tactically meaningful pressing environments, the framework aims to capture coordinated settled high pressing rather than generic defensive activity.

4.2 Pressing Run Detection

Within valid settled pressing contexts, the framework identifies individual **pressing runs** based on sustained defensive movement directed toward the current or anticipated possession target. Because pressing behaviour is not directly observable within tracking data, the framework operationalizes pressing through a combination of movement intensity, directional pursuit, spatial closing behaviour, and temporal persistence.

The resulting pressing runs represent the fundamental behavioural units used later in the construction of larger pressing sequences.

4.2.1 Movement Intensity

A primary characteristic of pressing behaviour is purposeful high-intensity defensive movement. To distinguish pressing from ordinary positional adjustment or low-intensity defensive positioning, the framework requires defenders to exceed a predefined movement speed threshold before a pressing run can be initiated.

Movement speed is evaluated continuously using player velocity information derived from the tracking data. Short isolated accelerations are not sufficient to trigger pressing detection. Instead, elevated movement intensity must persist across multiple consecutive frames in order to satisfy the initial behavioural requirements for pressing.

This persistence condition reduces false positives caused by momentary movement fluctuations or unrelated defensive repositioning.

4.2.2 Directional Pursuit

In addition to movement intensity, the framework evaluates whether the defender is actively moving toward the current or anticipated possession target. This is achieved by comparing the defender's movement vector with the directional vector connecting the defender to the target player.

Directional alignment is assessed over a short temporal window in order to reduce sensitivity to frame-level positional noise and rapid short-term movement variation. Defensive movement is only considered valid pressing behaviour when sufficient alignment exists between the player's movement direction and the target direction.

By incorporating directional pursuit, the framework distinguishes purposeful pressure application from unrelated high-speed movement occurring elsewhere on the pitch.

4.2.3 Forward-Oriented Defensive Engagement

The framework additionally incorporates a directional constraint designed to exclude defensive retreat movement toward a team's own goal. This distinction is important because high-intensity defensive recovery runs may otherwise resemble pressing behaviour from a purely movement-based perspective.

To address this issue, pressing runs are only considered valid when defensive movement reflects forward-oriented engagement consistent with active pressure on opposition possession. Retreating defensive movement associated with fallback positioning or defensive recovery is therefore excluded from pressing classification.

This constraint improves the tactical interpretability of detected pressing behaviour and helps distinguish organized high pressing from deeper defensive repositioning.

4.2.4 Distance-Closing Behaviour

A valid pressing action must additionally involve meaningful reduction in distance between the defender and the possession target over time. This requirement ensures that pressing behaviour reflects genuine defensive engagement rather than simply directional movement in the general vicinity of the ball.

The framework therefore evaluates whether the defender is actively closing spatial distance to the target player across the evaluated movement window. Pressing runs are only maintained when measurable distance reduction is observed consistently alongside movement intensity and directional alignment conditions.

This behavioural condition further improves robustness by filtering out movement patterns unrelated to active defensive pressure. The degree of defender-target proximity achieved during sustained pressure additionally contributes to later sequence evaluation procedures introduced in Section 4.4.

4.2.5 Temporal Persistence Conditions

Because football movement is inherently dynamic and noisy at frame level, the framework incorporates temporal persistence conditions before confirming a pressing run. All behavioural criteria must remain satisfied across a minimum sequence of consecutive frames in order for pressing behaviour to be formally established.

Once initiated, short interruptions in movement intensity, directional alignment, or distance-closing behaviour are tolerated within a predefined temporal margin. This prevents brief fluctuations during realistic football movement from unnecessarily fragmenting continuous defensive pressure into multiple short pressing runs.

During grounded passing phases, the framework additionally incorporates anticipated possession logic introduced in Chapter 3. When the ball is temporarily in transit between players, the anticipated receiver becomes the temporary target of defensive pressure, allowing pressing runs to remain tactically continuous across normal passing exchanges.

The resulting pressing runs provide a robust behavioural representation of sustained defensive engagement directed toward opposition possession. These runs subsequently form the basis for the sequence construction procedures introduced in the following section.

4.3 Pressing Sequence Construction

Individual pressing runs identified through the behavioural procedures described in the previous section are aggregated into broader **pressing sequences** representing sustained phases of coordinated defensive pressure. Modelling pressing at sequence level allows the framework to capture the temporal continuity of defensive behaviour rather than treating pressure as a collection of isolated frame-level actions.

The resulting sequences form the primary analytical units used later for tactical interpretation, outcome evaluation, and press scoring.

4.3.1 Sequence Initialization

A pressing sequence is initiated when at least one valid pressing run is detected within an eligible settled pressing context. Once initiated, the sequence remains active while the framework continues to observe temporally connected defensive pressure directed toward opposition possession.

Each sequence is associated with a unique temporal interval defined by its starting and ending frames. The framework additionally records contextual attributes including involved players, sequence duration, possession state, and eventual termination outcome.

Because multiple defenders may participate simultaneously or sequentially within the same pressing phase, individual pressing runs are treated as components of a larger coordinated defensive sequence rather than as fully independent actions.

4.3.2 Continuity During Pass Transitions

Football possession is inherently continuous, and defensive pressure often persists across passing exchanges even when the ball is temporarily in transit between players. To preserve this continuity, the framework incorporates the anticipated possession procedure introduced earlier during preprocessing.

When no immediate ball carrier is identified during grounded passing phases, the anticipated receiving player becomes the temporary target of defensive pressure. This allows active pressing runs to remain connected across short pass transitions without artificially terminating the broader pressing sequence.

By maintaining continuity during normal passing exchanges, the framework better reflects the dynamic and coordinated nature of high pressing during structured opposition build-up.

4.3.3 Timeout and Grace Mechanisms

Short interruptions in detected pressing behaviour may occur naturally during coordinated defensive movement, positional adjustment, or brief fluctuations in player speed and directional alignment. To prevent excessive fragmentation of pressing sequences, the framework incorporates several temporal tolerance mechanisms.

Once a sequence is active, brief periods without confirmed pressing activity are tolerated within a predefined timeout window. Similarly, minor interruptions caused by short passing exchanges or temporary tracking instability do not immediately terminate the sequence.

These grace mechanisms allow the framework to preserve tactically continuous defensive behaviour while remaining robust to frame-level noise and short-term movement variability. Without such tolerance conditions, sustained pressing phases could fragment into multiple artificially short sequences that do not reflect realistic football behaviour.

4.3.4 Sequence Termination Logic

A pressing sequence terminates when the framework determines that meaningful defensive pressure is no longer being sustained or when the surrounding tactical context becomes invalid.

Several termination conditions are incorporated within the framework. Sequences may end due to:

- possession changes,
- progression beyond the pressing zone,
- prolonged absence of valid pressing activity,
- aerial or unstable ball states,
- stoppages,
- or out-of-play events.

Termination may additionally occur when the attacking team successfully bypasses the press and progresses into deeper attacking areas where defensive engagement no longer represents high pressing behaviour.

Once terminated, the pressing sequence is stored together with its associated contextual and behavioural information for later outcome classification and evaluation.

The resulting sequence construction framework provides a temporally continuous representation of settled high pressing that captures coordinated defensive behaviour across multiple movement phases and passing exchanges.

4.4 Outcome Classification and Press Scoring

Beyond detecting the presence of defensive pressure, the framework additionally evaluates the tactical consequences of each pressing sequence. This evaluative component is important because not all pressing sequences are equally effective. Some sequences successfully disrupt opposition build-up or recover possession, while others fail to meaningfully constrain attacking progression.

To capture these differences, the framework classifies pressing outcomes and assigns each sequence a composite press score reflecting both behavioural intensity and tactical effectiveness.

4.4.1 Pressing Outcome Categories

Each completed pressing sequence is assigned an outcome category based on the tactical consequences observed immediately following sequence termination. Outcome classification is determined using synchronized event annotations together with contextual tracking information.

Several distinct outcome categories are incorporated within the framework. These include:

- clean possession regains,
- loose-ball possession regains,
- forced backward progression,
- forced sideways progression,
- forced clearances,
- forced out-of-play situations,
- neutral outcomes,
- successful opposition progression beyond the press,
- and situations resulting in dangerous attacking advancement or shots.

Possession regains are differentiated according to the stability of the recovery situation, distinguishing between clearly controlled regains and more unstable loose-ball recoveries. Similarly, out-of-play situations are evaluated contextually depending on whether the resulting restart favours the pressing or attacking team. Forced clearances and constrained territorial retreat are additionally treated as tactically positive outcomes because they may disrupt structured opposition build-up despite not producing immediate possession recovery.

4.4.2 Composite Press Score Construction

In addition to categorical outcome classification, each pressing sequence is assigned a continuous press score designed to support comparative evaluation across sequences.

The scoring framework combines multiple weighted sequence characteristics into a single evaluative measure. The primary component of the score is the tactical outcome associated with the pressing sequence, as successful disruption or possession recovery generally represents more effective defensive behaviour than sustained movement activity alone.

Additional behavioural components are also incorporated within the evaluation process. These include sequence duration and the degree of defender-target proximity achieved during active pressure. Longer and more persistent pressing phases may reflect sustained defensive coordination, while closer defensive engagement may indicate more direct and effective pressure application.

To preserve interpretability, outcome-related components are weighted more heavily than secondary behavioural measures. This ensures that tactically meaningful defensive consequences remain the dominant determinant of sequence evaluation while still allowing sustained and spatially effective pressure behaviour to contribute positively to the final score.

This resulting scoring framework allows pressing sequences to be compared continuously rather than solely through discrete categorical labels.

4.4.3 Interpretation of Press Scores

The press score is intended as an interpretative evaluation measure rather than an absolute indicator of tactical quality. Higher scores generally correspond to pressing sequences that apply sustained defensive pressure while successfully constraining or disrupting opposition possession.

However, the framework does not assume that all successful defensive actions are identical in tactical value. For example, a short pressing sequence resulting in immediate possession recovery may receive a higher evaluation than a longer sequence that merely delays attacking progression without significantly disrupting possession.

Similarly, neutral or unsuccessful sequences may still contain meaningful defensive behaviour despite producing limited tactical impact. The scoring framework therefore provides a relative representation of pressing effectiveness rather than a definitive measure of defensive quality.

By combining behavioural detection, sequence construction, outcome classification, and continuous evaluation, the framework provides a comprehensive representation of settled high pressing as both a dynamic tactical process and an evaluative analytical construct.

Chapter 5

Results

5.1 Overview of Detected Pressing Sequences

Application of the proposed framework to the selected match produced a structured set of detected pressing sequences representing sustained phases of settled high pressing. The resulting sequences provide the foundation for subsequent spatial, temporal, and evaluative analysis.

Across the analysed match, the framework detected 148 pressing sequences comprising a total of 13,939 active pressing frames. Sequence durations varied substantially, ranging from short isolated pressure phases to longer periods of sustained coordinated defensive engagement. This variation reflects the dynamic nature of pressing behaviour during structured opposition build-up.

The detected sequences exhibited substantial variation in duration and temporal continuity, ranging from short isolated pressure phases to longer periods of sustained defensive engagement. While many sequences aligned closely with tactically meaningful pressing behaviour observed within the available match footage, some sequences were also affected by unstable contextual states and sequence-boundary ambiguity.

Overall, the framework identified pressing behaviour continuously throughout multiple phases of the match rather than concentrating exclusively around isolated events. The temporal continuity of the detected sequences suggests that the framework successfully captures broader periods of coordinated defensive activity rather than short-lived frame-level movement fluctuations.

Table 5.1 summarizes the principal descriptive characteristics of the detected pressing sequences.

Table 5.1: Descriptive Summary of Detected Pressing Sequences

Metric	**Value**
Total pressing sequences	148
Total active pressing frames	13,939
Mean sequence duration	3.15
Standard deviation of duration	3.34
Minimum sequence duration	0.00
Maximum sequence duration	23.60

The descriptive statistics demonstrate that the framework is capable of identifying repeated and temporally sustained pressing behaviour within a professional football match context. At the same time, qualitative inspection of selected sequences revealed that the framework remains sensitive to contextual ambiguity and parameter selection in certain situations. Although many detected sequences corresponded closely to tactically meaningful pressing behaviour observed in the available match footage, some false positives, false negatives, and imprecise sequence boundaries were also observed. These limitations are discussed in greater detail in Chapter 6.

The following sections examine the detected sequences in greater detail by analysing their spatial distribution, temporal characteristics, tactical outcomes, and representative sequence behaviour.

5.2 Spatial and Temporal Characteristics of Pressing

Beyond identifying individual pressing sequences, the framework additionally captures broader spatial and temporal variation in defensive pressure throughout the analysed match. Examination of these patterns provides insight into how pressing behaviour differed between teams, evolved across match periods, and fluctuated over time.

5.2.1 Spatial Distribution of Pressing Sequences

For visualization consistency, all pressing sequence start locations were directionally normalized such that the pressing team is always oriented from left to right across the pitch. Consequently, the team associated with each subplot is interpreted as defending the left side of the pitch while applying pressure toward the opposition goal on the right. This normalization allows direct comparison of pressing behaviour between teams and across both halves of play independent of the original match orientation.

Figure 5.1 illustrates the normalized spatial origins of detected pressing sequences for both teams across each half of play. Because the framework explicitly restricts sequence detection to predefined pressing contexts, the figure should not be interpreted as demonstrating unrestricted spatial distribution across the entire pitch. Instead, the visualization is primarily useful for

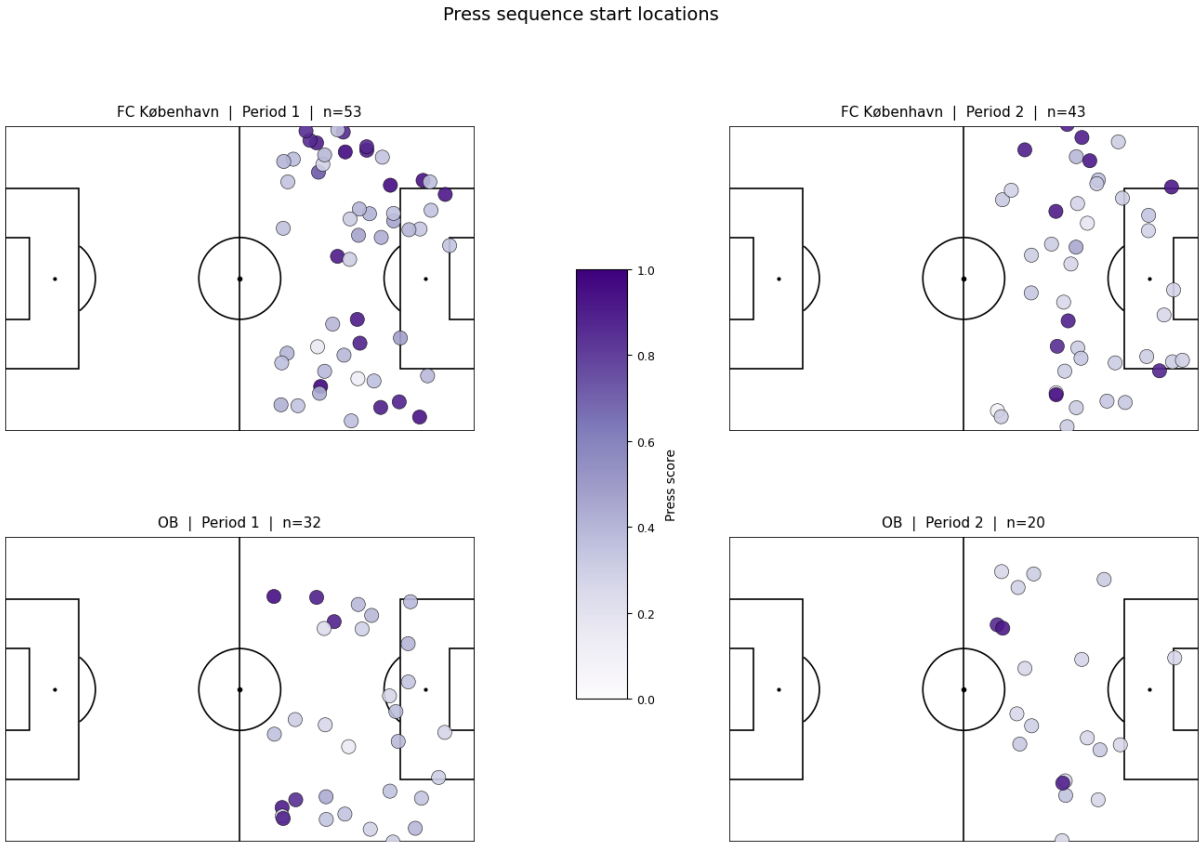


Figure 5.1: Press sequence start locations

examining relative differences in pressing behaviour between teams and match periods within the valid analytical regions defined by the framework.

Several observable differences emerged between the two teams. Certain periods exhibited denser concentrations of detected sequences, suggesting more sustained or aggressive defensive engagement during opposition build-up phases. Other periods displayed more dispersed or limited sequence distributions, reflecting comparatively reduced pressing activity or shorter-lived defensive pressure phases.

Differences were also observable between the first and second halves. These variations likely reflect changing tactical conditions, match dynamics, fatigue effects, and evolving game-state behaviour throughout the match. In particular, some halves exhibited noticeably greater concentrations of higher-scoring sequences, indicating periods in which pressing behaviour more consistently disrupted or constrained opposition progression.

Although the visualization does not provide definitive tactical conclusions from a single match sample, it demonstrates that the framework captures meaningful variation in pressing behaviour across teams and match periods rather than producing spatially uniform or randomly distributed outputs.

5.2.2 Temporal Evolution of Pressing Intensity

The aggregated press score profiles additionally exhibited substantial temporal variation throughout the match, reflecting fluctuations in the relative intensity and effectiveness of detected press-

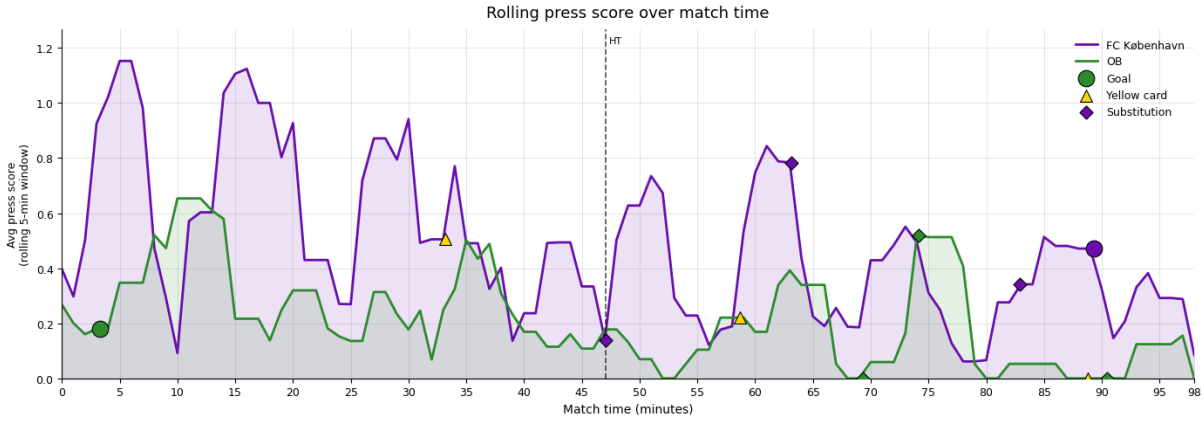


Figure 5.2: Rolling press score over match time

ing behaviour across different phases of play. Rather than occurring uniformly over time, pressing activity appeared in clustered phases characterized by repeated or sustained defensive engagement.

Figure 5.2 illustrates the temporal evolution of aggregated press score intensity across both halves of play. Rather than representing simple sequence frequency, the visualization reflects the relative evaluative strength of detected pressing behaviour over time by combining sequence occurrence with the composite press scoring framework introduced earlier.

Several periods of elevated press score intensity were observed in which sustained or tactically effective defensive pressure occurred repeatedly across short temporal intervals. These periods were interspersed with quieter phases characterized by comparatively limited or less effective pressing behaviour.

The temporal variation observed across the match likely reflects changing tactical priorities, possession structure, and match-state dynamics. Certain phases exhibited sustained periods of higher press score intensity associated with repeated or tactically effective defensive engagement, while other phases showed comparatively fragmented or lower-impact pressing behaviour.

Differences between the two halves were also evident in the density and continuity of detected sequences. In some periods, pressing activity appeared more fragmented and intermittent, whereas other phases exhibited longer and more continuous defensive engagement. These fluctuations support the importance of modelling pressing as a temporally continuous sequence-based process rather than as isolated frame-level actions.

Overall, the spatial and temporal analyses demonstrate that the framework captures structured variation in settled high pressing behaviour across different teams, match phases, and periods of play.

5.3 Pressing Outcomes and Effectiveness

Beyond identifying sustained phases of defensive pressure, the framework additionally evaluates the tactical consequences associated with each detected pressing sequence.

5.3.1 Distribution of Pressing Outcomes

Table 5.2 summarizes the distribution of outcome categories assigned to detected pressing sequences across the analysed match.

Table 5.2: Distribution of Pressing Outcomes

Outcome Category	**Count**	**Percentage**
Neutral outcome	65	43.92
Forced sideways progression	37	25
Clean possession regain	36	24.32
Press bypassed	8	5.41
Forced clearance	1	0.68
Forced out-of-play win	1	0.68
Forced out-of-play loss	0	0
Loose-ball regain	0	0
Forced backward progression	0	0
Shot conceded	0	0

Neutral outcomes represented the largest outcome category, indicating that many pressing phases temporarily constrained opposition progression without immediately producing possession recovery or complete attacking escape. Forced sideways progression and clean possession regains also occurred frequently, suggesting that the framework captured both indirect territorial disruption and direct defensive success.

Bypassed presses occurred comparatively infrequently relative to the total number of detected sequences. This indicates that many detected pressing phases maintained at least partial defensive control even when possession was not recovered directly.

Certain theoretically defined outcome categories occurred infrequently or were not identified robustly within the available match annotations. This reflects limitations in event-data richness and contextual recoverability, particularly for outcomes involving unstable possession transitions or ambiguous directional progression.

5.3.2 Press Score Analysis

The framework additionally generates a continuous press score for each detected sequence based on weighted tactical outcomes together with sequence duration and defender-target proximity during active pressure.

Observed press scores varied substantially throughout the match. The resulting press scores exhibited substantial variation across detected sequences, ranging from 0.087 to 0.977 with a standard deviation of 0.259. Sequences associated with clean possession regains or sustained territorial disruption generally received higher evaluations, while bypassed sequences typically received lower scores.

Importantly, sequence duration alone did not consistently correspond to stronger evaluations. Several shorter sequences received relatively high press scores due to immediate tactical disruption or possession recovery, whereas some longer sequences produced comparatively limited evaluative impact despite sustained defensive movement activity.

The rolling press score profiles additionally exhibited clear temporal variation across the match. Periods of elevated press score intensity frequently corresponded to sustained or tactically effective defensive engagement, while quieter phases reflected comparatively fragmented or lower-impact pressing behaviour.

Although the scoring framework depends on subjective weighting choices and contextual assumptions, the resulting evaluations generally aligned with qualitative observations from the available match footage. Overall, the press scoring framework provides a useful interpretative layer for comparing detected pressing sequences while preserving the contextual and continuous nature of settled high pressing behaviour.

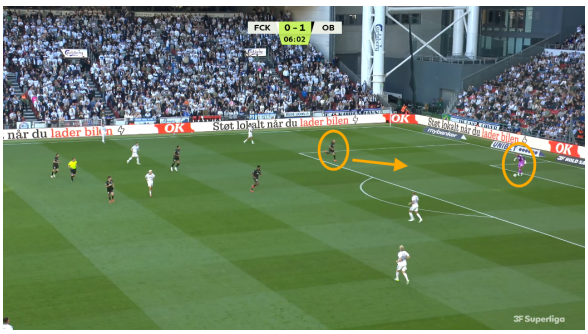
5.4 Case-Based Sequence Analysis

To further examine the behaviour of the proposed framework, two detected pressing sequences were analysed qualitatively. Both sequences resulted in a forced sideways outcome and received relatively similar press scores, but differed substantially in temporal structure and sequence continuity.

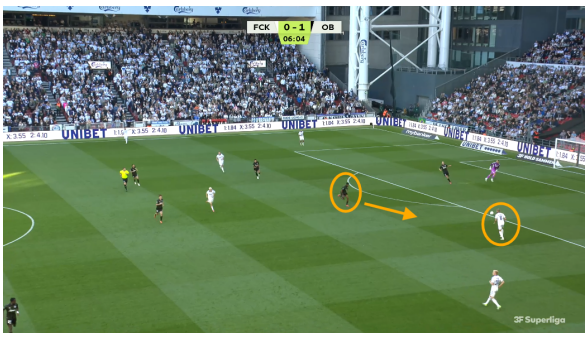
5.4.1 Example of Coherent Pressing Sequence Detection

The first example illustrates a short-duration pressing sequence lasting approximately 4.8 seconds. The sequence begins with an OB player applying pressure to the FCK goalkeeper immediately following a back pass. The goalkeeper subsequently passes to a nearby teammate who also comes under defensive pressure from a second OB player.

The sequence continues naturally across the pass transition before terminating once the next ball carrier receives no meaningful pressure. The resulting duration aligns closely with the visually observed period of coordinated defensive engagement and demonstrates coherent sequence initiation, continuity, and termination behaviour.



(a) Initial goalkeeper pressure



(b) Secondary pressure following pass transition



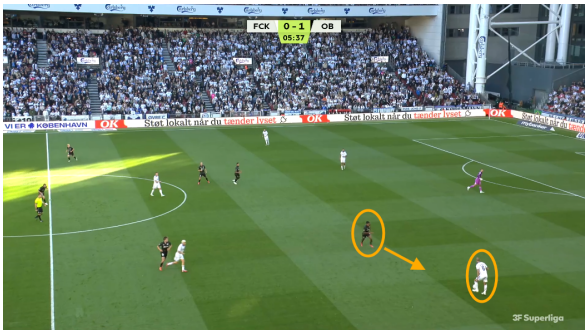
(c) Sequence termination after pressure release

Figure 5.3: Example of coherent pressing sequence detection

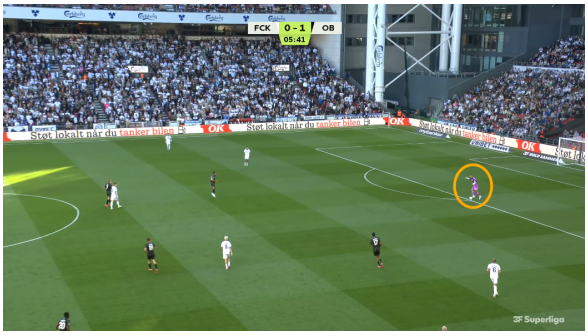
5.4.2 Example of Extended Sequence Continuity

The second example illustrates a longer 13-second pressing sequence exhibiting extended continuity persistence. The sequence begins plausibly, with pressure applied immediately after an FCK outfield player receives possession from the goalkeeper. However, after the ball is returned to the goalkeeper, active pressure temporarily disappears while the detected sequence itself continues.

The sequence subsequently persists across additional circulation phases before pressure is re-established later in possession. Although the framework maintains contextual continuity throughout this period, the resulting sequence likely contains multiple tactically distinct pressure phases that may have been merged into a single extended sequence.



(a) Initial pressure trigger



(b) Continuity persistence after temporary pressure loss

Figure 5.4: Example of extended sequence continuity

Together, these examples demonstrate both the interpretability of the proposed sequence-detection framework and the inherent difficulty of discretizing continuous pressing behaviour into temporally bounded analytical sequences.

Chapter 6

Discussion

6.1 Interpretation of Findings

The results presented in Chapter 5 demonstrate that the proposed framework is capable of identifying tactically plausible settled high-pressing sequences from synchronized event and tracking data. The detected sequences exhibited meaningful spatial and temporal variation throughout the analysed match, while the qualitative case-study analysis further demonstrated that the framework frequently captured coherent periods of coordinated defensive engagement.

The findings additionally reinforce the importance of modelling pressing as a continuous and context-dependent tactical process rather than as isolated defensive actions occurring at individual frames (Forcher et al., 2024; Herold et al., 2022). Several detected sequences developed progressively across multiple passes and player interactions, illustrating how defensive pressure may persist dynamically throughout opposition build-up phases. The contextual sequence-based formulation therefore provided a more realistic representation of pressing behaviour than approaches relying solely on instantaneous proximity or frame-level defensive events (Bekkers, 2024; Merckx et al., 2021).

The outcome distributions and press score evaluations further highlighted the multifaceted nature of pressing effectiveness. While clean possession regains represented some of the most valuable defensive outcomes, many detected sequences instead produced more indirect forms of tactical success such as constrained circulation or forced territorial retreat. This supports the broader interpretation that successful pressing cannot be evaluated exclusively through ball recovery events alone (Merckx et al., 2021; Robberechts, 2019).

At the same time, the results also demonstrated the inherent difficulty of discretizing continuous tactical behaviour into temporally bounded analytical sequences. Certain examples exhibited ambiguity regarding sequence continuity and termination, particularly in situations involving temporary reductions in active pressure followed by later defensive re-engagement. These observations highlight the sensitivity of pressing analysis to contextual assumptions, temporal thresholds, and sequence boundary definitions (Calabuig et al., 2024; Gudmundsson & Horton, 2017).

Overall, the findings suggest that the proposed framework provides a useful and interpretable foundation for contextual pressing analysis while simultaneously illustrating the com-

plexity and ambiguity inherent in modelling collective defensive behaviour in football.

6.2 Framework Strengths

6.2.1 Contextual Sequence-Based Modelling

One of the primary strengths of the proposed framework is its treatment of pressing as a continuous and context-dependent tactical process rather than as a collection of isolated defensive actions (Forcher et al., 2024; Herold et al., 2022). By modelling pressure sequences across temporally connected player interactions, the framework was able to capture sustained defensive engagement developing progressively throughout opposition build-up phases.

This sequence-based formulation produced more tactically interpretable outputs than approaches relying exclusively on instantaneous frame-level proximity measurements (Bekkers, 2024; Merckx et al., 2021). The qualitative sequence analysis additionally demonstrated that the framework frequently identified coherent initiation, continuation, and termination behaviour across multiple pass transitions and defensive interactions.

6.2.2 Integration of Event and Tracking Data

The integration of synchronized event and tracking data additionally provided an important methodological strength (Gudmundsson & Horton, 2017; Van Roy et al., 2023). Tracking data enabled detailed spatial and temporal modelling of player positioning and defensive pressure behaviour, while event annotations supplied contextual information regarding possession, passing actions, and tactical outcomes.

The combined use of both data sources allowed the framework to incorporate richer contextual reasoning throughout both sequence detection and evaluative analysis. In particular, event-derived information enabled the classification of pressing outcomes beyond simple possession recovery events, supporting a broader interpretation of defensive effectiveness.

6.2.3 Interpretable Evaluative Framework

A further strength of the framework lies in its flexible and interpretable evaluative structure. Rather than evaluating pressing exclusively through binary regain-versus-failure outcomes, the framework incorporated multiple categories of tactical consequence including constrained circulation, territorial retreat, and progression disruption (Merckx et al., 2021; Robberechts, 2019).

The composite press score additionally combined tactical outcomes with behavioural characteristics such as sequence duration and defender-target proximity, allowing different aspects of defensive pressure to contribute to the overall evaluation. Although inherently dependent on contextual assumptions and weighting choices, the resulting scoring framework provided a useful interpretative layer for comparing detected pressing sequences while preserving the continuous nature of collective defensive behaviour.

6.3 Limitations

6.3.1 Sequence Boundary Ambiguity

One of the primary limitations of the proposed framework concerns the difficulty of defining clear temporal boundaries for pressing sequences. Football pressing behaviour is inherently continuous and context-dependent, making it difficult to determine precisely when a sequence should begin or terminate (Calabuig et al., 2024; Forcher et al., 2024). Although the implemented continuity logic enabled the framework to capture sustained defensive engagement across multiple player interactions, certain situations exhibited ambiguity regarding whether temporary reductions in active pressure represented genuine sequence termination or short transitional interruptions within a larger pressing phase.

This occasionally resulted in extended sequences containing multiple tactically distinct pressure phases. In particular, prolonged circulation periods with intermittent defensive engagement sometimes produced sequence continuity beyond what may be considered a single coherent pressing action from a tactical perspective. These observations highlight the sensitivity of sequence-based modelling to timeout thresholds and contextual continuity assumptions.

Additional evidence of sequence-boundary instability was observed through the detection of several extremely short sequences lasting less than one second, including some with durations approaching zero. These fragmented detections likely reflect transient pressure triggers or unstable continuity transitions that did not correspond to tactically meaningful sustained pressing phases. This further illustrates the difficulty of defining robust temporal boundaries for continuous defensive behaviour.

6.3.2 Parameter Sensitivity

The framework additionally depends on several manually selected parameters and contextual assumptions. Press detection behaviour is influenced by choices relating to defensive pressure radius, sequence timeout duration, pressing-zone definition, and the weighting structure used within the composite press scoring framework.

Small adjustments to these parameters may substantially influence both the number and structure of detected pressing sequences. Similarly, the relative importance assigned to tactical outcomes, sequence duration, and defender-target proximity directly affects resulting score evaluations. This sensitivity reflects the broader difficulty of defining football pressing behaviour using universally accepted quantitative criteria, as tactical interpretations of pressure frequently vary across teams, analysts, and competitive contexts (Bekkers, 2024; Herold et al., 2022).

6.3.3 Dependence on Event-Data Richness

The evaluative capabilities of the framework are additionally constrained by the richness and granularity of the available event annotations. Certain tactical outcomes, particularly those involving unstable possession transitions or ambiguous loose-ball situations, proved difficult to identify robustly from the available data.

The absence of explicit contextual event labels also reduced the framework's ability to identify certain tactically meaningful pressing triggers. In particular, the dataset did not contain a dedicated back-pass event type despite back passes frequently acting as important triggers for coordinated pressing behaviour in real match situations. As a result, some pressing initiations could only be inferred indirectly through spatial and temporal player behaviour rather than through explicit event-based contextual information.

These limitations highlight the dependence of contextual pressing analysis on both accurate spatial tracking and sufficiently expressive event-data representation (Gudmundsson & Horton, 2017; Van Roy et al., 2023).

6.3.4 Limited Scope of Evaluation

The framework was evaluated using a single professional football match, limiting the broader generalizability of the presented findings. Although the selected case study provided sufficient variation to examine multiple forms of pressing behaviour and sequence continuity, different tactical systems, competitive contexts, and team styles may produce substantially different defensive dynamics.

Furthermore, the evaluation process primarily relied on qualitative tactical interpretation rather than large-scale quantitative benchmarking or manual expert annotation. While this approach aligned with the exploratory and interpretative objectives of the thesis, broader validation across multiple matches and tactical contexts would be necessary to assess the robustness and consistency of the framework more comprehensively.

6.4 Future Work

6.4.1 Adaptive Sequence Modelling

One potential direction for future work involves the development of more adaptive sequence continuity and termination logic. The current framework relies on fixed timeout thresholds and pressure persistence assumptions, which occasionally produced fragmented short-duration detections or extended sequences containing multiple tactically distinct pressure phases.

Future implementations could instead incorporate context-aware temporal modelling in which sequence continuation behaviour adapts dynamically to factors such as ball progression speed, local player density, or possession circulation structure (Calabuig et al., 2024; Forcher et al.,

2024). Such approaches may improve the robustness of sequence boundary detection while preserving the continuous nature of pressing behaviour.

6.4.2 Expanded Multi-Match Validation

The framework would additionally benefit from evaluation across a larger and more diverse collection of matches. Different teams, tactical systems, and competitive contexts may produce substantially different pressing structures and defensive behaviours.

Broader multi-match validation would therefore allow more comprehensive assessment of the framework's robustness, consistency, and tactical generalizability across varying football contexts.

6.4.3 Richer Contextual Integration

Future improvements could also incorporate richer contextual football information within both sequence detection and evaluative analysis. In particular, explicit identification of tactical pressing triggers such as back passes may improve the framework's ability to recognize coordinated defensive initiation phases more reliably.

Additional contextual features including formation structure, passing-lane accessibility, off-ball positioning, and local defensive compactness may further strengthen the tactical interpretability of detected pressing sequences (Fernandez & Bornn, 2018; Higgins et al., 2023).

6.4.4 Data-Driven Parameter Optimization

The current framework relies primarily on manually selected parameter values and scoring weights informed by tactical reasoning and iterative qualitative evaluation. Future work could instead explore more systematic calibration approaches using manually annotated pressing sequences or expert-labelled tactical datasets.

Such approaches may improve parameter robustness and reduce sensitivity to manually selected thresholds while preserving the interpretability of the underlying sequence-based framework.

6.5 Chapter Summary

This chapter discussed the broader implications, strengths, limitations, and future development possibilities associated with the proposed pressing-analysis framework. The findings demonstrated that contextual sequence-based modelling can provide tactically interpretable representations of settled high-pressing behaviour while simultaneously highlighting the inherent ambiguity involved in discretizing continuous defensive actions into temporally bounded analytical sequences.

Although the framework exhibited limitations relating to sequence continuity, parameter sensitivity, and event-data richness, the results nevertheless suggest that the integration of synchronized tracking and event data provides a useful foundation for contextual football pressing analysis. Future improvements involving adaptive temporal modelling, richer contextual integration, and broader multi-match validation may further strengthen the robustness and interpretability of the proposed approach.

Chapter 7

Conclusion

This thesis presented a contextual framework for the detection and evaluation of settled high-pressing sequences in professional football using synchronized event and tracking data. The proposed approach combined spatial pressure modelling, temporal sequence continuity, contextual possession analysis, and outcome-based evaluation to construct tactically interpretable representations of collective defensive behaviour.

Unlike frame-isolated approaches to pressure detection, the framework modelled pressing as a continuous and evolving tactical process developing across multiple player interactions and pass transitions. The integration of tracking and event data additionally enabled richer contextual reasoning throughout both sequence detection and evaluative analysis. Pressing effectiveness was therefore examined not only through direct possession recovery, but also through broader forms of tactical disruption such as constrained circulation and forced territorial retreat.

The resulting analyses demonstrated that the framework was generally capable of identifying tactically plausible pressing sequences exhibiting meaningful spatial and temporal variation throughout the analysed match. Qualitative case-study evaluation further showed that the framework frequently captured coherent sequence initiation, continuation, and termination behaviour. At the same time, the results also highlighted the inherent complexity of discretizing continuous defensive behaviour into temporally bounded analytical sequences. In particular, issues relating to sequence fragmentation, continuity persistence, parameter sensitivity, and event-data limitations demonstrated that contextual pressing analysis remains strongly dependent on modelling assumptions and tactical interpretation.

Despite these limitations, the proposed framework provides a useful and interpretable foundation for contextual football pressing analysis. More broadly, the thesis illustrates both the potential and the methodological difficulty of modelling collective defensive behaviour within complex dynamic team-sport environments.

References

Aközbek, K. C. (2019). *Finding patterns: Assessing defensive pressure in football using machine learning* [Doctoral dissertation, Tilburg University].

Bekkers, J. (2024). Pressing intensity: An intuitive measure for pressing in soccer. *arXiv preprint arXiv:2501.04712*.

Bialkowski, A., Lucey, P., Carr, P., Yue, Y., Sridharan, S., & Matthews, I. (2014). Large-scale analysis of soccer matches using spatiotemporal tracking data. *2014 IEEE international conference on data mining*, 725–730.

Calabuig, J. M., Catalán, C., García-Raffi, L. M., & Sánchez-Pérez, E. A. (2024). A mathematical model to study defensive metrics in football: Individual, collective and game pressures. *Mathematics*, *12*(23), 3854.

Fernandez, J., & Bornn, L. (2018). Wide open spaces: A statistical technique for measuring space creation in professional soccer. *Sloan sports analytics conference*, *2018*.

Forcher, L., Forcher, L., Altmann, S., Jekauc, D., & Kempe, M. (2024). The keys of pressing to gain the ball–characteristics of defensive pressure in elite soccer using tracking data. *Science and Medicine in Football*, *8*(2), 161–169.

Freitas, R., Volossovitch, A., Almeida, C. H., & Vleck, V. (2023). Elite-level defensive performance in football: A systematic review. *German Journal of Exercise and Sport Research*, *53*(4), 458–470.

Gudmundsson, J., & Horton, M. (2017). Spatio-temporal analysis of team sports. *ACM Computing Surveys (CSUR)*, *50*(2), 1–34.

Herold, M., Hecksteden, A., Radke, D., Goes, F., Nopp, S., Meyer, T., & Kempe, M. (2022). Off-ball behavior in association football: A data-driven model to measure changes in individual defensive pressure. *Journal of sports sciences*, *40*(12), 1412–1425.

Higgins, L., Galla, T., Prestidge, B., & Wyatt, T. (2023). Measuring the pitch control of professional football players using spatiotemporal tracking data. *Journal of Physics: Complexity*, *4*(2), 025008.

Linke, D., Link, D., & Lames, M. (2020). Football-specific validity of TRACAB's optical video tracking systems. *PloS one*, *15*(3), e0230179.

Low, B., Boas, G. V., Meyer, L., Lizaso, E., Hoitz, F., Leite, N., & Gonçalves, B. (2018). Exploring the effects of deep-defending vs high-press on footballers' tactical behaviour, physical and physiological performance: A pilot study. *Motriz: Revista de Educação Física*, *24*(02), e1018171.

Low, B., Rein, R., Raabe, D., Schwab, S., & Memmert, D. (2021). The porous high-press? An experimental approach investigating tactical behaviours from two pressing strategies in football. *Journal of sports sciences*, *39*(19), 2199–2210.

Merckx, S., Robberechts, P., Euvrard, Y., & Davis, J. (2021). Measuring the effectiveness of pressing in soccer. *Proceedings of the 8th Workshop on Machine Learning and Data Mining for Sports Analytics*.

Rein, R., & Memmert, D. (2016). Big data and tactical analysis in elite soccer: Future challenges and opportunities for sports science. *SpringerPlus*, *5*(1), 1410.

Robberechts, P. (2019). Valuing the art of pressing. *StatsBomb Innovation in Football Conference*, *11*.

Spearman, W., Basye, A., Dick, G., Hotovy, R., & Pop, P. (2017). Physics-based modeling of pass probabilities in soccer. *Proceeding of the 11th MIT Sloan Sports Analytics Conference*, *1*.

Van Roy, M., Cascioli, L., & Davis, J. (2023). ETSY: A rule-based approach to event and tracking data synchronization. *International workshop on machine learning and data mining for sports analytics*, 11–23.